



ENTO**FARM**

# WASTE-TO-RESOURCE CONVERSION AS A MODEL FOR CIRCULAR ECONOMY

<https://www.ento.farm/>



## MODULE OVERVIEW

The linear economic model (take-make-dispose) still dominates many production systems, leading to growing waste, pollution, and pressure on natural resources. Large amounts of materials such as food, plastics, paper, and textiles are discarded after a single use, even though many could be reused or recovered.

Waste-to-resource conversion is a key idea of the circular economy, which aims to keep materials in use longer and turn waste into valuable resources through solutions such as reuse, recycling, and composting.

This module helps students connect everyday waste streams with sustainable practices and responsible resource use.

## TARGET GROUP

**Age Range:** 12–18 years

**Subject Links:** STEM, Biology, Chemistry, Environmental Science, Civic Education, Economics

## LEARNING OBJECTIVES

By the end of this module, learners will:

**Knowledge:** Understand waste-to-resource conversion, the circular economy, and the main principles of the EU waste hierarchy.

**Skills:** Sort waste into streams, propose simple circular solutions, and design a small action plan to reduce and transform waste.

**Attitudes:** View waste as a potential resource, develop responsible consumption habits, and value cooperation and community action for sustainability.



SCAN ME

# Who are we?

## About the Project

This module has been developed within the framework of an **Erasmus+ proposal** focusing on **sustainable agriculture, circular economy, and innovative learning** approaches in STEM education.

The project brings together educational institutions, universities, and organizations from **Greece, Bulgaria and North Macedonia** to develop engaging educational materials for secondary school students. Its main objective is to help young people better understand how sustainable practices can address environmental challenges while **encouraging interest in STEM education and career**.

A central element of the project is the EntoFARM educational concept, which introduces students to alternative and sustainable food production systems such as insect farming and other circular resource management practices. Through these examples, students explore how biological processes, technological innovation, and responsible resource use can contribute to more sustainable food systems.

The educational materials produced in the project aim to support teachers in introducing sustainability and circular economy concepts through interactive and interdisciplinary learning activities.



## Project Consortium:

- Fraud Line – Greece (Project Coordinator)
- University of Thessaly – Greece
- University of Patras – Greece
- 1st EPAL Volou – Greece
- Professional Agricultural School “Buzema” – Bulgaria
- Professional Forum for Education – Bulgaria
- Association for Internationalization of Education and Science – North Macedonia

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## KEY CONCEPTS & THEMATIC CONTENT

### Why waste-to-resource matters?



Waste creates environmental problems when it is landfilled or unmanaged. A major issue with landfilled organic waste is methane release during decomposition.

A circular economy aims to:

- eliminate waste and pollution by design
- circulate products and materials at their highest value
- regenerate nature

# CORE PRINCIPLES OF WASTE-TO-RESOURCE CONVERSION



## Waste Hierarchy (Best Order of Actions)

The waste hierarchy is a simple rule:

1. Prevent (don't create waste)
2. Prepare for reuse (repair, clean, refill)
3. Recycle (turn materials into new materials)
4. Recover (energy recovery in controlled systems)
5. Dispose (landfill as last option)



## Source Separation (Sorting at the Start)

Waste-to-resource works best when waste is separated into streams:

- organic (food, garden waste)
- paper/cardboard
- plastics
- metals
- glass
- textiles
- e-waste (special handling only)

Sorting reduces contamination and improves recycling quality.



## Biological vs Technical Cycles

- **Biological cycle:** materials that can safely return to nature (food waste, plant waste). Key processes include composting and anaerobic digestion.
- **Technical cycle:** materials that should stay in use through reuse, repair, refurbishment, and recycling (metals, plastics, electronics).



# CORE PRINCIPLES OF WASTE-TO-RESOURCE CONVERSION



## Life-Cycle Thinking

Life-cycle thinking asks: "What impacts happen from start to finish?"

It helps learners compare options fairly (transport, energy, emissions, waste). This supports better choices, not just "recycling everything."



## Resource Recovery and Value Creation

Waste-to-resource conversion aims to produce:

- materials (recycled products)
- nutrients (compost/digestate for soil)
- energy (biogas in controlled digesters)

# MAIN WASTE-TO-RESOURCE PATHWAYS



## Waste streams and practical conversion pathways

Food & garden waste

**Prevention:** better planning, smaller portions

**Composting:** soil conditioner, nutrient return

**Anaerobic digestion:** biogas + digestate

## Paper & cardboard

**Reuse:** storage, crafts, packaging

**Recycling:** new paper products (needs clean streams)



## Plastics

**Reduce:** avoid single-use where possible

**Reuse:** refill systems, durable products

**Recycling:** depends on polymer type and local systems



# MAIN WASTE-TO-RESOURCE PATHWAYS



## Metals & glass

**Reuse and recycling** are highly effective  
(good material value)

## Textiles

- Repair, swap, reuse first
- Recycling where systems exist (often harder than people think)



## Biomass residues (wood, pruning waste)

### Composting (if suitable)

**Biochar (advanced):** carbon-rich soil material made by heating biomass in low oxygen

# REAL WORLD APPLICATIONS

## 1. City Organics to Biogas

Many cities use separate collection of organic waste and convert it into biogas and digestate through anaerobic digestion.

## 2. Industrial Symbiosis: “One Company’s Waste = Another’s Resource”

Kalundborg Symbiosis is a well-known example where companies exchange energy, water, and by-products to reduce waste and costs.

## 3. Organic Waste as a Resource in School Projects

Schools can transform cafeteria and garden waste into compost and use it in a school garden, showing a local circular loop.

# MISCONCEPTIONS TO ADDRESS

| Misconception                                  | Clarification                                                                    |
|------------------------------------------------|----------------------------------------------------------------------------------|
| Recycling is always the best solution          | Prevention and reuse come first in the waste hierarchy                           |
| Biodegradable” means it can be thrown anywhere | Many biodegradable items need specific conditions; littering still harms nature. |
| Composting is dirty and unsafe                 | With correct materials and hygiene, composting is safe and educational.          |
| Waste-to-energy means “problem solved.”        | Energy recovery is lower priority than prevention, reuse, and recycling.         |
| All plastics can be recycled easily.           | Plastic recycling depends on type, contamination, and local facilities.          |

## **CASE STUDY:** FROM BINS TO BENEFITS — A SCHOOL WASTE-TO- RESOURCE PLAN

### **Background/ Setting:**

A school wants to reduce waste and improve sustainability learning.

School cafeteria, classrooms, schoolyard, local community partners.



### **Scenario:**

Students map the school's waste streams for one week, then design a plan to:

- prevent waste (especially food waste)
- improve sorting and recycling quality
- start or improve composting
- connect with a local recycling/composting service



### **Purpose:**

Show how different waste streams can be prevented, reused, or converted into resources.



### **Reflection questions:**

- Which waste stream is the biggest in our school?
- What is the best action according to the waste hierarchy?
- What can we realistically do within one month?
- How can we include the whole school community?

# ASSESSMENT & EVALUATION

## Quiz: Waste-to-Resource Conversion

### Multiple Choice

1. **The best first step in the waste hierarchy is:**

- a. Landfill
- b. Recycling
- c. Prevention
- d. Incineration

2. **Which option is usually better than recycling?**

- a. Reuse/repair
- b. Landfill
- c. Littering
- d. Burning waste

3. **Anaerobic digestion happens:**

- a. With oxygen
- b. Without oxygen
- c. Only in oceans
- d. Only in landfills

4. **The main products of anaerobic digestion are:**

- a. Plastic and glass
- b. Biogas and digestate
- c. Coal and oil
- d. Metal and paper

5. **Composting is mainly used for:**

- a. Metals
- b. Organic waste
- c. Batteries
- d. Glass

6. **The circular economy aims to:**

- a. Increase waste exports
- b. Keep materials in use and regenerate nature
- c. Focus only on landfills
- d. Stop all production

7. **Which is an example of industrial symbiosis?**

- a. One factory buys new raw materials only
- b. Companies exchange by-products and energy as resources
- c. Burning mixed waste in open air
- d. Throwing food waste into rivers

8. **Which waste stream belongs to the biological cycle?**

- a. Food waste
- b. Aluminum cans
- c. Glass bottles
- d. Electronics



# ASSESSMENT & EVALUATION

## Quiz: Waste-to-Resource Conversion

### Multiple Choice

9. **Why is landfilling organic waste a major problem?**

- a. It makes compost faster
- b. It releases methane during decomposition
- c. It produces clean water
- d. It increases recycling rates

10. **A school waste map is mainly used to:**

- a. Blame students
- b. Identify waste streams and plan better actions
- c. Increase landfill use
- d. Mix waste to save time

# ACTIVITIES

## Activity 1: Design Your Own Waste-to-Resource System (School Version)

**Duration:** 60–90 minutes

**Objective:** Learners design a simple circular model using their school's waste streams.

### Activities:

Quick recap: waste hierarchy + circular loops (10 min)

Group work: identify 3 waste streams (food, paper, plastics) (15 min)

For each stream, define:

- Inputs (what waste, where from)
- Process (prevention, reuse, recycling, composting)
- Outputs (resource or avoided waste)

Create a visual model (poster or digital) (20–30 min)

Group presentations + peer feedback (15–20 min)

Expected Outputs:

- One “waste-to-resource flow” diagram per group
- A short explanation of benefits and challenges

Skills Developed:

- Systems thinking, teamwork, communication, practical planning

# ACTIVITIES

## Activity 2: Mini Waste Audit + Action Plan

**Duration:** 60–90 minutes

**Objective:** Learners measure a small part of school waste and create an improvement plan.

### Activities:

Choose one location (cafeteria or a hallway) and observe waste for one day (teacher-organized, safe method)

Categorize waste into streams using labels/cards (no hazardous items)

Optional: measure volume/weight (simple scale)

Discuss results:

- What can be prevented?
- What can be reused?
- What can be separated better for recycling/compost?

Create a 1-month action plan:

- One behavior change (example: “water refill bottles”)
- One system change (example: “clear bin labels”)
- One community action (example: “compost for school garden”)

Expected Outputs:

- A short waste audit summary (counts or simple chart)
- A realistic 1-month improvement plan

Skills Developed:

- Data observation, responsibility, problem-solving, civic action

# ADDITIONAL RESOURCES

## Links:

- European Commission overview of the Waste Framework Directive and waste hierarchy ([https://environment.ec.europa.eu/topics/waste-and-recycling/waste-framework-directive\\_en?](https://environment.ec.europa.eu/topics/waste-and-recycling/waste-framework-directive_en?))
- Ellen MacArthur Foundation circular economy overview and butterfly diagram (<https://www.ellenmacarthurfoundation.org/topics/circular-economy-introduction/overview?>)
- Anaerobic digestion outputs (biogas + digestate) overview (<https://docs.nrel.gov/docs/fy22osti/81676.pdf?>)
- EntoFARM project application (toolkit scope includes waste reduction, resource management, and waste-to-resource conversion).
- Anaerobic digestion outputs and explanation (biogas + digestate). (<https://docs.nrel.gov/docs/fy22osti/81676.pdf?>)
- European Commission Landfill methane risk from biodegradable waste. ([https://environment.ec.europa.eu/topics/waste-and-recycling/landfill-waste\\_en?](https://environment.ec.europa.eu/topics/waste-and-recycling/landfill-waste_en?))
- World Bank What a Waste 2.0 (global waste management overview). (<https://datatopics.worldbank.org/what-a-waste/?>)



# TEACHER REFLECTION AND ADAPTATION NOTES

This module has been designed as a flexible educational resource that can be adapted to different classroom contexts, student backgrounds, and educational levels.

Teachers are encouraged to reflect on how the activities and content can be adjusted to best support the needs of their students.

## Reflection Questions for Teachers

Before or after implementing the module, teachers may consider the following questions:

- Did the activities engage students with different interests and learning styles?
- Were all students encouraged to participate actively in discussions and group work?
- Did the module help students understand the connection between science, technology, and sustainability?
- Were examples and explanations accessible to students with different levels of prior knowledge?

## Accessibility Considerations

To ensure that the module is accessible to all learners, teachers may consider:

- simplifying technical terminology when necessary
- providing visual aids, diagrams, or short videos
- allowing students to express their understanding through different formats (presentations, posters, discussions, or written reflections)
- offering additional guidance or scaffolding for students who may need more support

## Supporting Girls' Engagement in STEM

Research shows that girls are more likely to engage with STEM when learning environments are collaborative and when scientific topics are linked to real-world challenges and social impact.

Teachers can therefore:

- encourage mixed-gender teamwork and equal participation
- highlight female role models in sustainability science and agriculture
- frame STEM challenges as problem-solving activities that benefit society and the environment
- ensure that classroom discussions give equal voice to all students

# TEACHER REFLECTION AND ADAPTATION NOTES

## Adapting the Module

Teachers are free to adapt the module by:

- adjusting the depth of scientific explanations
- modifying activities to fit available time or resources
- connecting the topic to local environmental or agricultural examples
- integrating the module with subjects such as biology, geography, environmental science, or technology

The goal is to ensure that the module remains relevant, inclusive, and meaningful for students in different educational settings.

## Do you have any questions?

Please let us know! We are happy to help.

## Contact us at:



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